Siddhartha Prasad

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I am a researcher focused on helping people write programs that behave as they intend. My research interests are informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

Education

Brown University Providence, RI *Jan 2022 – present*
Ph.D. in Computer Science

Tufts University Medford, MA *May 2016*
B.S. in Computer Science and Mathematics
Summa cum Laude; High Thesis Honors

Employment

Apple Seattle, WA

Research Intern · Human-Centered Machine Learning *May – Sept 2023*

Explored how AI and formal-methods techniques can support early childhood education.

Microsoft Redmond, WA

Software Engineer II · Applied AI *April 2018 – Sept 2021*

- Developer on the Azure Cognitive Services team, which lets developers embed AI into apps, bots, and websites.
- Built a containerized framework that allowed AI models to be run portably across operating systems and architectures.

Software Engineer · Developer Ecosystem Platform *Aug 2016 – March 2018*

- Developed APIs and features for XAML, a UI language used to build Universal Windows apps.
- Designed XAML APIs for input modalities (keyboard, gamepad, ink), accessibility, and cross-platform language support.

Software Engineering Intern · Developer Ecosystem *May – Aug 2015*

INRIA Saclay, France

Research Intern · Parsifal *June – July 2014*

Research on correctness certificates for first-order term rewriting.

Publications

Diagramming Program Values by Spatial Refinement
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi
PLDI 2026

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages
Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi
ECOOP 2026

hS: Speculative Script Reordering at Subprocess Granularity
Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, *Siddhartha Prasad*, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis
[To appear in] *OSDI 2026*

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
CAV 2025 **[Distinguished Paper]**

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
ECOOP 2025 **[Distinguished Paper; Distinguished Artifact]**

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic
Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte
FM 2024

ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading
Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro
IDC 2024

Forge: A Tool and Language for Teaching Formal Methods
Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi
OOPSLA 2024

Conceptual Mutation Testing for Student Programming Misconceptions
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
Programming 2024

Generating Programs Trivially: Student Use of Large Language Models
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
CompEd 2023

Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions
Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi
Koli Calling 2022

Large-Scale Intelligent Microservices
Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman
IEEE BigData 2020

Teaching

Teaching Assistant

2022–2026	CSCI 1710 — Logic for Systems	Brown University
2025	CSCI 1730 — Programming Languages	Brown University
2015–2016	CS 105 — Programming Languages	Tufts University
2014	CS 160 — Algorithms	Tufts University

Service

2026 Artifact Reviewer, *The Art, Science, and Engineering of Programming*

Affiliations & Certifications

Sheridan Teaching Seminar Certificate	Brown University
Tau Beta Pi, The Engineering Honor Society	Tufts University